

Arunesh Kumar Lal

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EDUCATION

Boston University Graduate School of Arts and Sciences
Master of Science in Computer Science(Data Centric Computing)

Boston, MA
Sep 2023 - Jan 2025

Manipal Institute of Technology
Bachelor of Technology in Electronics,Instrumentation and Control

Manipal,India
Aug 2016 - May 2020

SKILLS

Data Science: Machine Learning, Deep Learning, Predictive Modeling, Statistical Analysis, Feature Engineering, Anomaly Detection, Data Wrangling, EDA

Programming & Development: Python, SQL, Java, JavaScript, TypeScript, PySpark, Data Structures, Algorithms

Big Data & Cloud Technologies: Spark, Hadoop, Kafka, AWS (Lambda, S3, EC2), GCP, Docker, Kubernetes

Visualization & AI Tools: Tableau, Power BI, TensorFlow, PyTorch, XGBoost, Scikit-learn

Database Management: PostgreSQL, MongoDB, MySQL, Microsoft SQL Server

Risk & Identity Solutions: Fraud Detection, Model Validation, Financial Data Analytics, A/B Testing

Business & Communication: Data Storytelling, Stakeholder Communication, Strategic Decision-Making

WORK EXPERIENCE

Boston University
Graduate Full Stack Teaching Assistant

Boston,MA
Aug 2024 – Dec 2024

- **Automated Grading System:** Developed and deployed an automated grading system reducing grading time by 75%.
- **Full-Stack Development Assistance:** Mentored 30+ students in full-stack development, leading hands-on sessions on front-end technologies such as HTML, CSS, and React, along with back-end development utilizing FastAPI.

ZF Digital Solutions
Application Developer Intern

Bengaluru, IN
May 2022 – July 2022

- **Cross-Functional Collaboration and A/B Testing:** Partnered with cross-functional teams to gather user feedback and implement A/B testing for prototype iterations, resulting in data-driven design improvements.
- **UI Design and Functionality Enhancements:** Enhanced the UI design and improved front-end functionality for a login app through iterative design changes informed by A/B testing, resulting in a 20% increase in user engagement.

KPMG Assurance and Consulting Services LLP
Risk Analyst

Bengaluru, IN
Jan 2020 – Dec 2020

- **Fraud Detection Optimization:** Identified fraud patterns using Excel,SQL and Power BI, **reducing financial losses by INR 100 million/quarter.**
- **Process Improvement and Data Analytics:** Conducted data analysis on returns data to identify process inefficiencies, implementing data-driven solutions that **saved INR 150 million per quarter** and improved inventory accuracy by 25%.
- **Quality Control Enhancement:**Implemented a **data-driven quality control system** that reduced product scrapping risk by **90%** by optimizing return handling processes.

UNIVERSITY PROJECTS

Epidemic Engine Data Pipeline

Feb2024 - Apr 2024

- **Scalable Data Pipeline Development:** Designed and implemented a scalable data pipeline using **Hadoop, Spark, Kafka**, and **PostgreSQL**, enabling efficient data ingestion, processing, and storage for epidemic data analysis.
- **Anomaly Detection with Machine Learning:** Applied **Random Forest** and **K-Means clustering** models to detect anomalies in epidemic data, achieving **93.7% accuracy** and improving risk prediction capabilities.

Kalshi Trading Automation Projects

Feb 2024 - Apr 2024

- **Automated Trading System Development:** Designed and implemented an **automated trading system** that leverages **climate forecasting** and **predictive modeling** to execute real-time trades with enhanced accuracy and efficiency.
- **Predictive Modeling and Real-Time Integration:** Built a **high-accuracy temperature forecasting model** for four major cities with **95% accuracy** using historical data and statistical techniques, and developed a **custom API** to automate real-time trades, improving risk-adjusted decision-making.

CERTIFICATIONS

AWS Certified: (AI Practitioner & Cloud Practitioner)

March 2025

Finance: Columbia University Financial Engineering and Risk Management

March 2025

Technology: IBM Data Science Specialization

March 2025